

Foundation (Jalapeno)

- Q1.** What is the decimal equivalent of the fraction $\frac{3}{4}$?
(A) 0.25
(B) 0.5
(C) 0.75
(D) 0.8
(E) 0.9
- Q2.** A space station is traveling at $\frac{3}{5}$ of the speed of light. What percent is this?
(A) 20
(B) 30
(C) 40
(D) 50
(E) 60
- Q3.** Which of the following rational numbers is the largest: $\frac{1}{2}$, 0.4, or $\frac{3}{5}$?
(A) $\frac{1}{2}$
(B) 0.4
(C) $\frac{3}{5}$
(D) They are equal
(E) None of the above
- Q4.** A sample of Moon rock weighs 2.5 kg. What fraction of a kilogram is 0.8 kg?
(A) $\frac{1}{4}$
(B) $\frac{1}{3}$
(C) $\frac{2}{5}$
(D) $\frac{3}{5}$
(E) $\frac{2}{3}$
- Q5.** Order the rational numbers $\frac{2}{3}$, 0.6, and $\frac{3}{4}$ from least to greatest.
(A) $\frac{2}{3}$, 0.6, $\frac{3}{4}$
(B) 0.6, $\frac{2}{3}$, $\frac{3}{4}$
(C) $\frac{3}{4}$, $\frac{2}{3}$, 0.6
(D) 0.6, $\frac{3}{4}$, $\frac{2}{3}$
(E) $\frac{2}{3}$, $\frac{3}{4}$, 0.6
- Q6.** The cost of launching a spacecraft is $\frac{3}{4}$ million dollars. What is this amount in decimal form?
(A) 0.25 million
(B) 0.5 million
(C) 0.75 million
(D) 0.8 million
(E) 0.9 million
- Q7.** A physics experiment measures a distance of 2.4 meters. What fraction of a meter is 0.8 meters?
(A) $\frac{1}{3}$
(B) $\frac{1}{4}$
(C) $\frac{1}{2}$
(D) $\frac{2}{3}$
(E) $\frac{3}{4}$
- Q8.** A spaceship travels $\frac{2}{3}$ of the distance to Mars. What percent of the distance is this?
(A) 20
(B) 30
(C) 40
(D) 50
(E) 66.67

Open Ended Questions (FRQ)

- Q9.** Plot the rational numbers $\frac{1}{2}$, 0.7, and $\frac{3}{4}$ on a number line. Then, determine which rational number is the largest and explain your reasoning.
- Q10.** A scientist measures the temperature of a sample to be -2.5 degrees Celsius. Convert this temperature to a fraction and then determine what percent this is of -5 degrees Celsius. Show all work and explain your reasoning.

Chef's Tips

- Remind students to show all work for free-response questions
- Encourage students to use number lines to visualize rational numbers
- Emphasize the importance of converting between fractions, decimals, and percents in real-world applications